

Is the auxiliary loss doing anything?

Gradient geometry of multi-term objectives, with applications to structural losses for precipitation super-resolution

Lecture notes for the `minkowski-loss` project

23 September 2026

Abstract

You train a network on $L_0 + \lambda L_k$ and want to know whether L_k took any part in shaping the result. This is harder than it looks. Loss *values* carry no information about influence. The obvious gradient diagnostic, comparing $\|\nabla L_0\|$ with $\lambda \|\nabla L_k\|$, is only meaningful far from convergence. Adaptive optimisers change the geometry in which “how much” is measured. These notes build the question up from first principles:

- one step of gradient descent;
- the equilibrium of the combined objective, via linear-response theory;
- stochastic dynamics, where the aux term’s drift competes with minibatch noise;
- Adam’s preconditioned geometry;
- statistical estimation from minibatches.

We then derive what the common weighting schemes (uncertainty weighting, GradNorm, MGDA, PCGrad, black-box search) actually optimise. Finally, we apply everything to the five structural losses of Study 1 and to the audit tool in `tools/gradient_audit.py`. We aim at understanding, not recipes. Each section ends with what it implies for the measurement.

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1 The question, stated carefully

Let $\theta \in \mathbb{R}^D$ be the parameters ($D \approx 3.1 \times 10^7$ for our U-Net). We train on

$$L(\theta) = L_0(\theta) + \lambda L_k(\theta), \quad (1)$$

where L_0 is the pixel MSE in log-normalised space, L_k is one of the structural losses (Minkowski, spectral, SSIM, wet area, optical flow), and $\lambda > 0$ is fixed after a warm-up. We write $\mathbf{g}_0 = \nabla L_0$ and $\mathbf{g}_k = \nabla L_k$ for the full-data gradients. A single minibatch B gives unbiased noisy estimates $\hat{\mathbf{g}}_0, \hat{\mathbf{g}}_k$.

“Is L_k active?” has three non-equivalent meanings, and it pays to keep them apart:

- (a) **Instantaneous.** Does the term change the direction of the update *now*? This is a local statement about $\mathbf{g}_0$ and $\lambda \mathbf{g}_k$ at one point.
- (b) **Cumulative.** Does the term change *where training ends up*? This is a statement about the solution θ_λ against θ_0 . It depends on curvature and on noise, not just on gradient sizes.
- (c) **Observable.** Does the term change a metric we care about (MAE, exceedance ratio, the value of L_k itself)? This is what the paper reports. It is a function of (b), seen through the metric.

The audit numbers map onto these as follows:

- the *share* r and the cosine address (a);
- the “own-loss” test and the λ -scaling predictions of §4 address (b) and (c);
- the MAE “activity check” addresses (c), but turns out to be a *second-order* and hence weak test (§4).

2 Why loss values carry no information

Proposition 1 (Gauge freedom of a loss term). *For any $c > 0$ and $a \in \mathbb{R}$, training on $L_0 + \lambda L_k$ and on $L_0 + (\lambda/c)(cL_k + a)$ produces identical gradients, and therefore identical trajectories under any first-order optimiser.*

Proof. $\nabla[(\lambda/c)(cL_k + a)] = \lambda \nabla L_k$. □

The value of L_k is thus defined only up to an affine map, and the value ratio $\lambda L_k / L_0$ is not a physical quantity.

- Our Minkowski loss takes values ~ 1 and our spectral loss $\sim 4 \times 10^{-3}$. That tells you nothing about which one steers more.
- The additive constant matters more than it seems. SSIM has an *unreachable floor* of 0.465 on our data (§8.2), so 93% of its value is a constant with zero gradient.

Any weighting rule built on values inherits this arbitrariness. We will see in §7.1 that uncertainty weighting is exactly such a rule.

The invariant objects are the gradients $\lambda \mathbf{g}_k$, and more precisely their action through the optimiser.

3 One step of gradient descent

Take one full-batch GD step $\Delta \theta = -\eta(\mathbf{g}_0 + \lambda \mathbf{g}_k)$. Let ψ be the angle between $\mathbf{g}_0$ and $\mathbf{g}_k$.

Definition 1 (Share and parity weight).

$$r = \frac{\lambda \|\mathbf{g}_k\|}{\|\mathbf{g}_0\|}, \quad \lambda^* = \frac{\|\mathbf{g}_0\|}{\|\mathbf{g}_k\|}, \quad \text{so } r = \lambda/\lambda^*. \quad (2)$$

Direction. The update is deflected away from $-\mathbf{g}_0$ by an angle φ with

$$\tan \varphi = \frac{r \sin \psi}{1 + r \cos \psi}. \quad (3)$$

Two numbers from the first audit, spectral at $r = 0.014$ and optical flow at $r = 0.007$, give $\varphi < 0.8^\circ$. The update is, for all practical purposes, the MSE update.

What happens to each loss. To first order in η ,

$$\Delta L_0 = -\eta \left(\|\mathbf{g}_0\|^2 + \lambda \langle \mathbf{g}_0, \mathbf{g}_k \rangle \right) = -\eta \|\mathbf{g}_0\|^2 (1 + r \cos \psi), \quad (4)$$

$$\Delta L_k = -\eta \left(\lambda \|\mathbf{g}_k\|^2 + \langle \mathbf{g}_0, \mathbf{g}_k \rangle \right) = -\eta \|\mathbf{g}_0\| \|\mathbf{g}_k\| (r + \cos \psi). \quad (5)$$

Equation (5) is the heart of the matter. The change in L_k has a *self* term r and a *bystander* term $\cos \psi$ (the aux loss moves because the MSE moves). The ratio self/bystander is $r/\cos \psi$. Three regimes follow:

- $r \ll |\cos \psi|$: L_k is a *bystander*. Whatever it does is inherited from the MSE. A model trained this way should score on L_k like a vanilla model trained for as long.
- $r \gg |\cos \psi|$: L_k drives its own descent.
- In high dimension, independent directions have $\cos \psi = O(D^{-1/2})$. A small measured $\cos \psi$ (Minkowski: -0.02 in the first audit) means that even a modest r is enough for the self term to dominate. *Small r is not the same as inactive; what matters is r relative to $|\cos \psi|$.*

Equation (4) shows the first-order cost to L_0 : only a *conflicting* aux gradient ($\cos \psi < 0$) slows the MSE. An orthogonal one costs nothing at first order. Its cost appears at second order, which is the subject of the next section.

Remark. λ^* is a *unit*, not an optimum. It is the weight at which the two terms have equal say in the instantaneous direction. Our Minkowski loss works at $r \approx 2$ –5 and games the metrics at $10\times$ its weight. Nothing forces the good λ to be near λ^* . Its use is to express every loss's weight in the same dimensionless unit, which is what a fair comparison needs.

4 Where training ends up: equilibrium and linear response

One step tells you about the direction. The solution is determined by a balance of forces. Physicists will recognise the structure: the auxiliary loss is an external field, the base loss is the potential, and the Hessian is the stiffness.

4.1 Perturbation theory around the base minimum

Let θ_0 be a non-degenerate local minimum of L_0 , with Hessian $H = \nabla^2 L_0(\theta_0) \succ 0$. Write $g_k = \nabla L_k(\theta_0)$. By the implicit function theorem the minimiser of $L_0 + \lambda L_k$ satisfies

$$\theta_\lambda = \theta_0 - \lambda H^{-1} g_k + O(\lambda^2). \quad (6)$$

Proposition 2 (Response of the two losses). *Define the susceptibility $\chi_k = g_k^\top H^{-1} g_k > 0$. Then*

$$L_k(\theta_\lambda) - L_k(\theta_0) = -\lambda \chi_k + O(\lambda^2), \quad (7)$$

$$L_0(\theta_\lambda) - L_0(\theta_0) = \frac{1}{2} \lambda^2 \chi_k + O(\lambda^3). \quad (8)$$

Eliminating λ , the local Pareto front is a parabola:

$$\Delta L_0 \approx \frac{(\Delta L_k)^2}{2\chi_k}. \quad (9)$$

Proof. Taylor-expand both losses around θ_0 using (6). For L_k the linear term is $g_k^\top \Delta\theta = -\lambda \chi_k$. For L_0 the gradient vanishes at θ_0 , leaving $\frac{1}{2} \Delta\theta^\top H \Delta\theta = \frac{1}{2} \lambda^2 \chi_k$. $\square$

Three consequences shape the whole audit.

1. The MAE activity check is a second-order test. The cost to the base loss grows like λ^2 , and the gain on the aux loss like λ . So at small λ a genuinely active term can improve its own objective noticeably while moving MSE or MAE imperceptibly. “ $\Delta\text{MAE} < 1\%$ means inactive” is therefore a sufficient signal only when combined with “and its own loss did not improve either”. The **own-loss test** (the trained model evaluated on its own L_k , against vanilla) is first order and strictly more sensitive. It is the primary evidence in the audit.

2. The right norm is H^{-1} , not Euclidean. χ_k weights the aux gradient by the *inverse* stiffness of the base loss.

- Directions in which the MSE is flat (small curvature) respond enormously to a small push. Directions in which it is stiff barely move.
- The Euclidean share r ignores this completely, so it is a proxy for (b) at best.
- This is also where reward hacking lives. The grid-aligned filaments of the $w = 10^{-3}$ Minkowski run (DECISIONS.md §2) are, almost by definition, a direction the MSE does not resist much and the Minkowski loss rewards.

3. At equilibrium the full gradients balance exactly. At a stationary point of $L_0 + \lambda L_k$ one has $g_0 = -\lambda g_k$. So

$$r = 1, \quad \cos \psi = -1 \quad (\text{full batch, at the combined optimum}). \quad (10)$$

This gives a clean diagnostic for a model trained *with* L_k : at its own checkpoint the noise-corrected share should approach 1 and the cosine -1 . The size of the deviation measures how far from equilibrium training stopped.

- Conversely, at the *vanilla* checkpoint $\mathbf{g}_0 \approx 0$ (it is a minimum of L_0). So $\lambda^* = \|\mathbf{g}_0\| / \|\mathbf{g}_k\| \rightarrow 0$ and $r \rightarrow \infty$ for *every* aux loss.
- The parity weight evaluated at the ERM solution is therefore not a well-posed quantity, contrary to the prescription in DECISIONS.md §9 and in Chang & Sapsis. What one measures there, with minibatches, is mostly the noise scale of $\hat{\mathbf{g}}_0$ (§5).
- The well-posed places to measure parity are the **initialisation**, or early training, where both gradients are signal-dominated. There the ratio governs the transient that decides which basin you enter.

4.2 The overparameterised caveat

With $D \gg$ (effective number of constraints), H is singular. The minimisers of L_0 form a manifold $\mathcal{M}$, and (6) holds only in the directions transverse to $\mathcal{M}$.

- Along $\mathcal{M}$, L_0 exerts no restoring force. An arbitrarily small λ then drives the solution, given enough time, to $\arg \min_{\mathcal{M}} L_k$: the aux loss acts as a tie-breaker at zero cost to L_0 .
- In that regime $\chi_k = \infty$ in the flat directions. MAE-based checks see nothing, while L_k can move by $O(1)$.
- Our U-Net does not interpolate the data (validation MSE is far from zero, and the targets are noisy), and training is finite. So we are in an intermediate regime: some flat directions, many stiff ones. It is one more reason to trust the own-loss test over the MAE check.

5 Noise: drift against diffusion

Minibatch gradients are $\hat{\mathbf{g}} = \boldsymbol{\mu} + \boldsymbol{\xi}$ with $\mathbb{E}\boldsymbol{\xi} = 0$ and $\text{Cov } \boldsymbol{\xi} = \Sigma/B$ (approximately, for batch size B). The aux term contributes a *drift* $\lambda\boldsymbol{\mu}_k$. The base loss contributes a restoring force plus diffusion. Whether the drift matters is a signal-to-noise question.

5.1 A one-dimensional model

Project the dynamics onto the unit vector $\mathbf{e} = \boldsymbol{\mu}_k / \|\boldsymbol{\mu}_k\|$. Near a minimum of L_0 with curvature $h = \mathbf{e}^\top H \mathbf{e}$ along $\mathbf{e}$, SGD with learning rate η is a discretised Ornstein–Uhlenbeck process:

$$x_{t+1} = x_t - \eta(h x_t + \lambda \|\boldsymbol{\mu}_k\|) + \eta \sigma_e \zeta_t, \quad \sigma_e^2 = \mathbf{e}^\top \Sigma \mathbf{e} / B, \quad \zeta_t \sim \mathcal{N}(0, 1). \quad (11)$$

For small ηh the stationary distribution has

$$\text{mean shift } m = -\frac{\lambda \|\boldsymbol{\mu}_k\|}{h}, \quad \text{spread } s = \sigma_e \sqrt{\frac{\eta}{2h}}, \quad \frac{|m|}{s} = \frac{\lambda \|\boldsymbol{\mu}_k\|}{\sigma_e} \sqrt{\frac{2}{\eta h}}. \quad (12)$$

The relaxation time is $1/(\eta h)$ steps. Three lessons:

- An aux term can shift the *mean* solution while being invisible in any single step, because $\lambda \|\boldsymbol{\mu}_k\| \ll \sigma_e$. A consistent drift accumulates; noise does not.
- Without restoring force ($h \rightarrow 0$, a flat direction) the drift wins after $T^* \approx (\sigma_e / (\lambda \|\boldsymbol{\mu}_k\|))^2$ steps.
- Learning-rate decay ($\eta \downarrow$, as ReduceLROnPlateau does) *raises* $|m|/s$. Late training is where small, consistent aux terms leave their mark.

5.2 What a per-batch norm measures

Lemma 1. $\mathbb{E} \|\hat{\mathbf{g}}\|^2 = \|\boldsymbol{\mu}\|^2 + \text{tr } \Sigma / B$.

Near a minimum of L_0 , $\|\boldsymbol{\mu}_0\| \rightarrow 0$ while $\text{tr } \Sigma_0 / B$ does not. A per-batch $\|\hat{\mathbf{g}}_0\|$ is then a measure of *noise*, and a parity ratio built from per-batch norms compares the aux signal with

the MSE noise. That is a meaningful comparison (it is $\lambda \|\mu_k\| / \sigma$ above, up to the $\sqrt{\eta h}$ factor), but it is not the parity of §3.

The first audit (`tools/gradient_audit_v0`) used per-batch norms at the vanilla checkpoint. Its “ λ^* ” values are therefore noise-referenced. The qualitative ordering is still informative, but the “70 $\times$ ” and “140 $\times$ ” factors should be read in that light.

5.3 Unbiased estimation

With N minibatch gradients $\hat{\mathbf{g}}^{(i)}$, mean $\bar{\mathbf{g}}$ and mean squared norm $S = \frac{1}{N} \sum_i \|\hat{\mathbf{g}}^{(i)}\|^2$:

$$\widehat{\|\mu\|^2} = \frac{N \|\bar{\mathbf{g}}\|^2 - S}{N - 1}, \quad \widehat{\langle \mu_0, \mu_k \rangle} = \frac{N \langle \bar{\mathbf{g}}_0, \bar{\mathbf{g}}_k \rangle - \frac{1}{N} \sum_i \langle \hat{\mathbf{g}}_0^{(i)}, \hat{\mathbf{g}}_k^{(i)} \rangle}{N - 1}. \quad (13)$$

Both follow from Lemma 1 and its bilinear analogue. The second one matters because $\hat{\mathbf{g}}_0$ and $\hat{\mathbf{g}}_k$ are computed on the *same* batch, so their noises are correlated. The naive inner product of mean gradients is biased by that cross-covariance, by $\text{tr} \Sigma_{0k} / (NB)$. So even for orthogonal signals the naive cosine is not centred on zero when N is small.

The tool reports:

- the noise-corrected norms and cosine;
- the *noise fraction* $1 - \widehat{\|\mu\|^2} / \text{median } \|\hat{\mathbf{g}}\|^2$;
- jackknife standard errors over K groups of batches.

When $\widehat{\|\mu_0\|^2} \leq 0$, the MSE signal is statistically indistinguishable from zero, and the tool says so instead of printing a meaningless ratio.

6 Adam changes the geometry

We train with AdamW. Its update is

$$\theta \leftarrow \theta(1 - \eta w) - \eta \frac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}} + \epsilon}}, \quad (14)$$

with $\hat{\mathbf{m}} \approx \mathbb{E} \hat{\mathbf{g}}$ and $\hat{\mathbf{v}} \approx \mathbb{E} \hat{\mathbf{g}}^{\odot 2}$ (coordinate-wise), and decoupled weight decay w .

Preconditioned share. Holding $\hat{\mathbf{v}}$ fixed, adding $\lambda \mathbf{g}_k$ changes the update by $-\eta \lambda P \mathbf{g}_k$ with $P = \text{diag}(1/(\sqrt{\hat{\mathbf{v}} + \epsilon}))$. The relevant share is therefore

$$r_P = \frac{\lambda \|P \mathbf{g}_k\|}{\|P \mathbf{g}_0\|}. \quad (15)$$

The tool computes it with $\hat{\mathbf{v}}$ taken from each checkpoint’s saved optimizer state.

- Near convergence $\hat{\mathbf{v}}$ is dominated by noise, so $P \approx \text{diag}(1/\sigma_i)$. Adam then measures each coordinate’s gradient *in units of its own noise*, and r_P is a signal-to-noise-weighted share.
- A loss whose gradient concentrates in coordinates where the MSE is quiet gets amplified. One spread over noisy coordinates gets suppressed.
- In the smoke test (4 batches, not to be quoted) the preconditioner cut SSIM’s share by almost an order of magnitude and wet area’s by seven. Euclidean and preconditioned shares can tell different stories.

Invariances. Adam is invariant to a global rescaling of the *total* loss (up to ϵ). It is *not* invariant to λ , since that changes the relative composition of $\hat{\mathbf{m}}$ and $\hat{\mathbf{v}}$. So “Adam normalises the gradients, so weights don’t matter” is wrong. They matter through the direction.

The AdamW fixed point is preconditioner-dependent. Stationarity requires $P\nabla L = -w\theta$, not $\nabla L = 0$. The linear response (6) becomes

$$\theta_\lambda - \theta_0 \approx -\lambda(PH + wI)^{-1}P\mathbf{g}_k \quad (P \text{ held fixed}), \quad (16)$$

which reduces to $-\lambda H^{-1}\mathbf{g}_k$ as $w \rightarrow 0$. The equilibrium itself therefore depends on the optimiser’s statistics. This is one more reason the Euclidean parity rule is only a first approximation.

Clipping and mixed precision.

- *Clipping.* Global-norm clipping at $c = 1$ rescales the total update. It changes the step size, not the share, and only when $\|\hat{\mathbf{g}}\| > 1$. The tool reports how often that happens.
- *fp16 underflow.* Training runs in fp16 autocast with a **GradScaler**. The scaled backward pass multiplies gradients by $S \sim 2^{16}$ before unscaling. A very small λ times small per-element aux gradients can nevertheless underflow in fp16 intermediates (the smallest subnormal is $\approx 6 \times 10^{-8}$). **This is a hypothesis, not a finding.** But for $\lambda = 4 \times 10^{-5}$ (optical flow) it is worth checking, because it would make a small weight even smaller in practice. The audit runs in fp32 and cannot see it. A direct test compares one fp16 and one fp32 backward on the same batch.

7 What the standard weighting schemes optimise

7.1 Uncertainty (homoscedastic) weighting

Kendall, Gal & Cipolla (2018) minimise

$$\mathcal{L}(\theta, \sigma) = \sum_k \frac{1}{2\sigma_k^2} L_k(\theta) + \log \sigma_k. \quad (17)$$

Proposition 3. *At the optimum over σ_k one has $\sigma_k^2 = L_k$. The effective weight is then $1/(2L_k)$, and the θ -gradient becomes*

$$\nabla_\theta \mathcal{L} = \sum_k \frac{\nabla L_k}{2L_k} = \frac{1}{2} \nabla \sum_k \log L_k. \quad (18)$$

Proof. $\partial_{\sigma_k} \mathcal{L} = -L_k/\sigma_k^3 + 1/\sigma_k = 0$. □

So uncertainty weighting equalises *relative* (logarithmic) progress across terms.

- It is invariant to rescaling $L_k \rightarrow cL_k$, which is good.
- It is *not* invariant to offsets. With an unreachable floor $L_k = f + \ell_k$, the weight is $1/(2(f + \ell_k))$. For SSIM, $f = 0.465$ dominates, so the floor sets the weight.
- It is a Gaussian-likelihood argument. None of our structural losses is a Gaussian negative log-likelihood, so the σ_k have no probabilistic meaning here.
- The weight is *learned* per loss. In a comparison of losses, that means each loss gets an uncontrolled, loss-specific λ , which is exactly what a fair comparison must avoid.

It is well suited to balancing the three Minkowski channels (area, perimeter, χ) against each other, since they are commensurable; that is where the project already uses it.

7.2 GradNorm

Chen et al. (2018) adapt $\lambda_k(t)$ so that the weighted gradient norms (on the last shared layer) track a common target, modulated by each task’s relative training rate. It is *dynamic parity*: it drives $r \rightarrow$ target throughout training. It solves the units problem. But it fixes the operating point at parity, whereas our experience (the Minkowski loss works at $r \approx 2$ –5 and hacks at $\approx 10\times$ that weight) says the right point is loss-specific.

7.3 MGDA and PCGrad

- *MGDA* (Sener & Koltun 2018) picks the min-norm point of the convex hull of $\{g_k\}$, which gives a common descent direction and converges to Pareto stationarity. It encodes no preference between objectives, and here we have a strong one: the MSE is primary.
- *PCGrad* (Yu et al. 2020) projects out conflicting components when $\cos \psi < 0$. With $\cos \psi \approx 0$, as measured for Minkowski, it does essentially nothing.

7.4 Black-box search (Optuna, Bayesian optimisation)

Treat λ as a hyperparameter and maximise a validation score. This is principled, with two caveats:

- *The result is the chosen metric.* Choosing it is the actual scientific decision, and it must not be any loss’s own objective.
- *The search is one-dimensional per loss and expensive.* A handful of bracketed runs covers it. Search pays off for joint spaces ($\lambda \times$ anneal $\times$ tail sampling), ideally on short proxy runs with pruning and an equal budget per loss.

7.5 What a fair comparison needs

Put every loss in the same dimensionless unit, then probe the response.

1. Measure the parity unit λ^* where it is well posed: at initialisation or early training (§4), Euclidean and Adam-preconditioned.
2. Train a fixed log-bracket around it, reaching above parity.
3. Evaluate at a fixed budget, on the last checkpoint.
4. Check the *signatures* of Proposition 2:
 - own-loss gain $\propto \lambda$;
 - base-loss cost $\propto \lambda^2$;
 - at the trained checkpoint, share $\rightarrow 1$ and cosine $\rightarrow -1$.

A loss that shows no linear own-loss gain across the bracket is genuinely inert, not merely under-weighted.

This is the protocol in DECISIONS.md §16, with the correction that parity is measured at initialisation rather than at the ERM solution.

8 Case studies: where each structural loss puts its gradient

The gradient of a loss is a field over pixels, $\partial L_k / \partial \hat{y}_p$, pulled back through the network. Its *support* (which pixels receive signal) explains most of the audit.

8.1 Minkowski: threshold-localised gradients and the pixel-mass bottleneck

The analytical loss replaces the indicator $\mathbb{K}[\hat{y}_p \geq u]$ by $s_p(u) = \sigma((\hat{y}_p - u)/\tau_u)$, with $\tau_u = \max(0.1u, \tau_{\min})$. Area, perimeter and Euler characteristic are smooth functionals of $\{s_p(u)\}$. Since $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ decays like $e^{-|z|}$, pixel p receives gradient from threshold u only if $|\hat{y}_p - u| \lesssim$ a few τ_u . The number of pixels in that band is

$$N_u \approx |D| \int_{u-c\tau_u}^{u+c\tau_u} f(y) dy \approx 2c|D|\tau_u f(u) \propto u f(u), \quad (19)$$

where f is the pixel-intensity density.

- For a heavy tail $f(u) \propto u^{-\alpha}$ with $\alpha > 1$, $N_u \propto u^{1-\alpha}$ decreases with u . The thresholds at 31–150 mm/h see almost no pixels in their band.

- That is the measured result that 83% of the gradient comes from $u \leq 0.28$ mm/h and ≈ 0 from $u \geq 31$ (DECISIONS.md §5), derived rather than observed.
- Widening τ (annealing) raises every band proportionally, so the *share* cannot change. That is the anneal-sweep result.
- The remedies that change N_u itself are tail-sample selection and threshold reweighting.

8.2 SSIM: an unreachable floor

Per window, $\text{SSIM} = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2) + \varepsilon}$, with $c_{1,2} = (k_{1,2}R)^2$, where R is the per-sample data range. The implementation also adds $\varepsilon = 10^{-6}$ to the denominator.

In an exactly dry window of both prediction and target, $\mu = \sigma = 0$, so $\text{SSIM} = c_1c_2/(c_1c_2 + \varepsilon)$. With $k_1 = 0.01$, $k_2 = 0.03$ and $R \leq 1$ in normalised units, $c_1c_2 \leq 9 \times 10^{-8}R^4 \ll \varepsilon$. Hence $\text{SSIM} \approx 0$ and the per-window loss $(1 - \text{SSIM})/2 \approx \frac{1}{2}$, *even for a perfect prediction*. The dataset-level floor is

$$L_{\text{SSIM}}^{\min} \approx \frac{1}{2} \times (\text{fraction of dry windows}) = 0.465 \Rightarrow \approx 93\% \text{ of windows dry.} \quad (20)$$

The gradient lives in the remaining $\approx 7\%$, over a dynamic range of $0.465 \rightarrow 0.5$. Two fixes:

- a fixed data range ($R = 1$ in normalised space) with $\varepsilon \ll c_1c_2$, which makes dry-on-dry windows score 1;
- or restricting the mean to windows that are wet in either field.

8.3 Spectral, wet area, optical flow

- **Spectral** (FFT-magnitude energy score) is translation-invariant, so its gradient is orthogonal to every translation of the prediction. It cannot penalise displacement. Its value is an L^1 over all rfft bins, dominated by high-wavenumber bins of nearly dry fields, which explains its tiny scale.
- **Wet area** is the single-threshold ($u = 0.1$ mm/h, $\tau = 0.05$) special case of the Minkowski area channel. Its gradient support is the drizzle band only, so it is blind to intensity. It can be “active” ($r \gg 1$) and still useless for extremes.
- **Optical flow** uses the coarse input as its teacher. That teacher is smooth with suppressed peaks, so the gradient points toward resembling the input.

9 Reading the audit

tools/gradient_audit.py reports, per checkpoint and loss:

Column	Definition	Answers
parity λ^*	$\widehat{\ \mu_0\ } / \widehat{\ \mu_k\ }$, eq. (13)	the unit of weight
share r	λ / λ^*	instantaneous influence (a)
per-batch λ^*	median of $\ \hat{g}_0\ / \ \hat{g}_k\ $	noise-referenced (the v0 number)
cos, cos corr.	naive and noise-corrected	bystander vs self, eq. (5)
Adam share	$\lambda \ \underline{P\bar{g}_k}\ / \ P\bar{g}_0\ $	influence on the real update
noise fraction	$1 - \ \mu\ ^2 / \text{med } \ \hat{g}\ ^2$	is this norm signal?
clip engaged	share of batches with $\ \hat{g}_0 + \lambda\hat{g}_k\ > 1$	does clipping act?
own vs vanilla	paired, bootstrap CI	first-order effect (b,c)

Checklist:

1. **Initialisation:** is r (Euclidean) small, and is it small relative to $|\cos \psi|$? That tells you whether the term could have shaped the early trajectory.
2. **Vanilla checkpoint:** expect $\|\widehat{\boldsymbol{\mu}}_0\| \approx 0$ and a noise fraction near 1 for the MSE. Parity is undefined there.
3. **Own checkpoint of a run trained with L_k :** does the share approach 1 and the cosine -1 ? If not, the run was stopped far from the combined equilibrium, or the term never mattered.
4. **Own-loss test:** does the trained model beat vanilla on its own L_k , with a CI that excludes 0? This is the decisive first-order evidence.
5. **Across a bracket** (after the M4 reruns): is the own-loss gain linear in λ and the MSE cost quadratic? That confirms the linear-response regime and locates where it breaks down (hacking).

10 Exercises

Exercise 1. Show that $\tan \varphi = r \sin \psi / (1 + r \cos \psi)$ in (3). For fixed $\psi > \pi/2$, how does φ behave as r grows, and at which r does the update stop decreasing L_0 ? What happens when $\cos \psi = -1$?

Exercise 2. Prove Lemma 1 and derive the unbiased estimators in (13). Why does the naive cosine of two mean gradients computed on the same batches not tend to 0 for independent signals when N is small?

Exercise 3. Using Proposition 2, show that if a run at λ improves its own loss by δ at an MSE cost ϵ , then a run at $\lambda/3$ should improve by $\approx \delta/3$ at a cost of $\approx \epsilon/9$. How would you use this to tell “inert” from “under-weighted” with three runs?

Exercise 4. In the OU model of §5, compute the number of steps after which the mean shift reaches $1 - e^{-1}$ of its stationary value. With $\eta = 5 \times 10^{-5}$, how large must h be for a 25-epoch run (about 4.5×10^5 steps at batch 128 on 2.3 M patches) to equilibrate?

Exercise 5. Show that uncertainty weighting applied to $L_k = f + \ell_k$ with a constant floor $f > 0$ gives effective weight $1/(2(f + \ell_k))$. For SSIM ($f = 0.465$, $\ell_k \in [0, 0.035]$), how much does the effective weight change over the whole achievable range of ℓ_k ? What does this say about what the learned σ_k can encode?

Brief solutions

1. Decompose $\lambda \mathbf{g}_k$ into components parallel and perpendicular to $\mathbf{g}_0$. For $\psi > \pi/2$, φ increases monotonically from 0 towards ψ as $r \rightarrow \infty$. It passes 90° at $r = -1/\cos \psi$, where the update has no component along $-\mathbf{g}_0$ and L_0 stops decreasing (eq. (4)). At $\cos \psi = -1$ and $r = 1$ the two gradients cancel exactly: that is the equilibrium of §4.
2. Expand $\mathbb{E} \|\bar{\mathbf{g}}\|^2 = \|\boldsymbol{\mu}\|^2 + \text{tr} \Sigma / (NB)$ and eliminate $\text{tr} \Sigma$ using $\mathbb{E} S$. Same-batch noise is correlated with covariance Σ_{0k} , which biases $\langle \bar{\mathbf{g}}_0, \bar{\mathbf{g}}_k \rangle$ by $\text{tr} \Sigma_{0k} / (NB)$.
3. Linear gain and quadratic cost. Three bracketed runs with no linear own-loss trend mean $\chi_k \approx 0$ (inert). A linear trend that is merely small means under-weighted.
4. $t \approx 1/(\eta h)$. Equilibration within 4.5×10^5 steps needs $h \gtrsim 1/(\eta \cdot 4.5 \times 10^5) \approx 0.044$ along $\mathbf{e}$. Flatter directions do not equilibrate within the run.
5. The weight varies by $< 8\%$ over the achievable range, so σ_k encodes essentially only the floor.

A Measurements, 23 September 2026

tools/gradient_audit.py with $N = 512$ batches of 128 patches (65,536 patches, drawn with the training sampler), $K = 8$ jackknife groups, anneal 0.1, fp32, GPU 1. Raw output: eval_results/gradient_audit/2026-09-23/. Weights λ are those the Study-1 runs were trained with.

A.1 Parity at initialisation (where it is well posed)

At initialisation the MSE gradient is almost pure signal (noise fraction 0.01), so λ^* is well defined.

loss	λ used	λ^*	$r = \lambda/\lambda^*$	$\cos \psi$
Minkowski	1×10^{-4}	1.05×10^{-3}	0.095	+0.78
spectral	5×10^{-4}	0.298	1.7×10^{-3}	+0.96
SSIM	1.6×10^{-4}	0.932	1.7×10^{-4}	-0.77
wet area	2×10^{-2}	2.00×10^{-2}	1.0	+0.85
optical flow	4×10^{-5}	1.09	3.7×10^{-5}	+0.79

At initialisation every aux gradient is strongly aligned with the MSE gradient ($|\cos \psi| \approx 0.8\text{--}0.96$), so every term starts in the bystander regime $r \ll |\cos \psi|$ of eq. (5). Even Minkowski starts $10\times$ below parity, and the term only takes over later, as the MSE gradient shrinks. Wet area is at parity and clips 64% of initial batches.

A.2 The vanilla checkpoint: parity is ill-posed there

At the converged vanilla checkpoint the MSE noise fraction is 0.96: the per-batch norm (3.0×10^{-4}) is almost all noise around a signal of 6.1×10^{-5} . The per-batch parity values of the first audit are reproduced (spectral 3.9×10^{-2} , v0: 3.6×10^{-2}). The noise-corrected values are 4–5 $\times$ smaller (spectral 9.2×10^{-3}). This is Lemma 1 in action. The “70 $\times$ below parity” of the first audit is $\approx 18\times$ after noise correction, and $\approx 600\times$ at initialisation. The factor depends entirely on where it is measured.

A.3 The equilibrium test at each run’s own checkpoint

For a term that trained actively, §4 predicts $r \rightarrow 1$ and $\cos \psi \rightarrow -1$.

run (own loss)	r	$\cos \psi$ (corr.)	Adam share	verdict
Minkowski	1.22	-0.986	1.10	at equilibrium: active
wet area	0.74	-0.967	1.03	at equilibrium: active
SSIM	0.157	+0.470	0.038	far from it: bystander
spectral	0.021	+0.915	0.022	far from it: bystander
optical flow	0.0066	-0.474	0.008	far from it: inert

The two active runs sit almost exactly on the equilibrium line, with Euclidean and Adam shares agreeing. The three others have shares $\ll 1$ and positive or weakly negative cosines. Their own gradient never balanced the MSE, and in two of them it still points the same way.

A.4 The own-loss test

Paired over the same 512 batches, with bootstrap 95% intervals, relative to vanilla: Minkowski -64.8% [-65.9, -63.8], wet area -13.9% [-15.8, -12.1], optical flow -2.6% [-6.4, +1.2], spectral +2.5% [+0.9, +4.4], SSIM +0.1% [+0.0, +0.2].

- Spectral and SSIM are *worse* on their own objective than the vanilla model.
- The 34-epoch vanilla model is also worse than the 64-epoch one on spectral (4.18 vs 4.10×10^{-3}). So the sign is explained by under-training at 25 epochs, not by the term.
- Both tests agree. Spectral, SSIM and optical flow were bystanders. Minkowski and wet area were active, and wet area’s activity is confined to the drizzle band, as its gradient support predicts.

A.5 Consequences for the reruns

- Use the *initialisation* λ^* as the unit. By it, the Study-1 weights were $\sim 10^{-1}$ (Minkowski), $\sim 10^{-3}$ (spectral), $\sim 10^{-4}$ (SSIM), $\sim 4 \times 10^{-5}$ (optical flow) and 1 (wet area) times parity. Minkowski became active only because $\|\mathbf{g}_0\|$ collapsed during training (A.3). The competing losses would need orders of magnitude more weight to follow the same route.
- A bracket expressed in init-parity units cannot be copied from Minkowski’s working point. Minkowski was active at $0.1\lambda_{\text{init}}^*$, yet that same fraction leaves the others inert, because their gradients are dense and aligned with the MSE. **The bracket should instead be placed so that the equilibrium test (A.3) is satisfied at the end of training,** and checked with the own-loss test.

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Author lists and venues of the machine-learning references are from memory. Check them before citing in the paper.